Dossier de définition
Iphigénie 60

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Contents

1 Goals & Experiments **2**
1.1 Goals . . . 2
1.2 Experiments . . . 2
1.3 Timeline . . . 3

2 Mechanical **4**
2.1 Initial design . . . 4
2.2 Systems . . . 5
2.2.1 Air frame . . . 5
2.2.2 Air brake . . . 5
2.2.3 Structural . . . 6
2.2.4 Reaction wheel . . . 6
2.2.5 Parachute opening mechanism . . . 6
2.3 Manufacturing . . . 6
2.4 Tests . . . 6

3 Electrical **7**

4 Team structure **8**
4.1 About . . . 8
4.2 Organization . . . 8

5 Questions **9**

1 Goals & Experiments

1.1 Goals

The design of the rocket is based on the goals set during its definition. The goals are:

1. Launch the rocket — we have not launched a rocket in the past two years, so this is our main priority.
2. Make the rocket modular (separate main and motor sections) — to allow reuse if manufacturing a new rocket is not feasible in future years, or if the team wants to optimize the existing rocket instead.
3. Build a base for the EuRoC competition — we also want to compete in other competitions, so this year we will create a base air-brake mechanism that can be reused in future years.
4. Optimize the rocket — using the knowledge of new members, we aim to optimize the rocket both aerodynamically and structurally, using CFD, FEA, and parametric and topology optimization.
5. Optimize and predict the rocket's trajectory — using an in-house simulator, we want to predict the trajectory as accurately as possible (verified against StabTraj) and stabilize the pitching and rolling dynamics using theoretical models, not only experimental ones (e.g. diameter-based static margin).
6. TODO: elec goals

1.2 Experiments

The following paragraphs describe the main experiments planned for the Iphigénie 60 rocket.

Air brake is the main experiment. The current design status and controller approach are detailed in section 2.2.2.

- *Goal:* characterize and validate the air brake's ability to control the rocket's drag so that it reaches a target apogee h_{target}, given that the motor's thrust is sized to overshoot h_{target} by default.
- *Parameters to study:* additional drag force (or drag coefficient) produced by the air brake as a function of deployment angle, the resulting apogee, and the vertical velocity/acceleration profile during the braking phase.
- *Measurement range and precision:* TODO:
- *Hypothesis:* without the air brake, the motor drives the rocket to overshoot h_{target}; by extending drag flaps during ascent, the air brake can bleed off the excess energy and bring the achieved apogee down to h_{target}.
- *Expected outcome:* the achieved apogee lands within a few percent of h_{target}, markedly closer than the uncontrolled (brakeless) apogee predicted in table 2.

Reaction wheel experiment. Background on our previous attempts is given in section 2.2.4.

- *Goal:* analyze whether a motor-driven reaction wheel can control the rocket's roll rate in flight, counteracting roll induced by fin misalignment and roll–pitch coupling.
- *Parameters to study:* roll rate over time, reaction-wheel motor torque/current, and the coupling between roll rate and pitch rate.
- *Measurement range and precision:* TODO:
- *Hypothesis:* a motor-driven reaction wheel can actively damp the rocket's roll rate independently of fin-induced torque, keeping it near zero during the camera-recording phase of flight despite roll–pitch coupling.
- *Expected outcome:* the roll rate measured with the reaction wheel active is substantially lower than in our previous (uncontrolled) attempts, giving steadier camera footage.

Telemetry TODO: elec experiments

Radar TODO: yahya

1.3 Timeline

The project is planned according to the following milestones:

- Design finished — end of December.
- Optimizations finished — mid-January.
- Manufacturing finished — end of February.

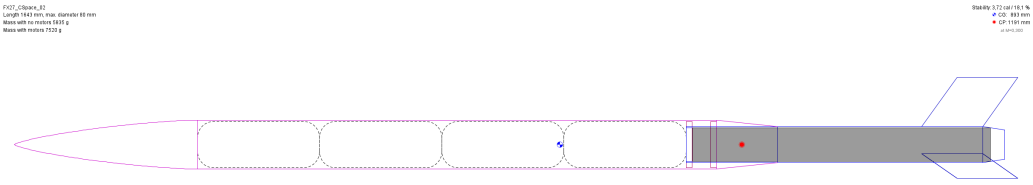
2 Mechanical

2.1 Initial design

Our current C'Space design consists of two configurations; the one we use will depend on the motor allocated to us and on our design goal of achieving a longer flight duration. We are still optimizing the rocket's drag and mass, so it is not yet clear which configuration will perform better. The two configurations are shown in figure 1.



(a) Pro75 configuration.



(b) Pro54 configuration.

Figure 1: The two rocket configurations under consideration.

Both configurations share a similar front part (main pipe) but differ in their motor pipes. From the nosecone back, the main pipe houses the parachute bay, the reaction wheel, the air brake, and the electronics bay. One of our goals was to make the rocket modular, which is illustrated here by its ability to accept different motor pipes. The coupling between the main and motor pipes will use the same interface for both the constant-diameter and the reducing-diameter (transition) configurations. The motors will seat flush with the motor pipe. However, this is less relevant for C'Space, since we will commit to one of the two configurations once the motor allocation is confirmed.

The overall dimensions of the rocket are summarized in table 1, listing the outer diameter and length of each pipe section, as well as the cumulative configuration length measured from the top of the rocket to the end of that pipe.

Table 1: Dimensions of the rocket's pipe sections.

Pipe	Outer diameter [mm]	Length [mm]	Configuration length [mm]
Main pipe	80	850	-
Motor pipe (Pro75)	80	534	1742
Motor pipe (Pro54)	59	486	1643

The current design goal is to maximize flight duration: keeping the air brake deployed for longer gives the controller more time to add drag and bleed off excess energy, allowing the rocket to hit its target apogee more precisely instead of overshooting. Current data shows that the Pro75 configuration has 5 s more flight time than the Pro54 configuration; we would therefore request motors in the following order of preference:

1. Pro75-3G Green
2. Pro54-5G Classic
3. Pro54-5G Green (our least preferred option, as it would give a short flight)

The current flight-data estimates are shown in table 2 and figure 2; everything will be verified with StabTraj.

These are the results of some initial simulations. We know the maximum apogee allowed for this competition is 3000 m, so we will further tune the rocket's mass and drag to bring the configurations within that range.

For the idealized configurations, the reported apogee corresponds to the estimated value when only the mass or only the drag is varied from its nominal configuration, in order to isolate the effect of each parameter.

The stability of these configurations has not yet been analyzed: the values used so far were chosen arbitrarily, only to give a workable stability margin. Static and dynamic stability analyses are planned in the near future, and everything will be verified with StabTraj.

Table 2: Preliminary flight-data estimates for the six configurations.

Configuration	m_{dry} [kg]	m_{wet} [kg]	apogee [m]	t_{apo} [s]	v_{max} [m/s]	a_{max} [m/s^2]	M_{max}	v_{rail} [m/s]
Pro54 (current)	6.530	7.520	2460.2	22.12	242.6	109.2	0.72	32.5
Pro75 (current)	7.905	9.977	3731.6	27.74	304.1	71.6	0.91	24.2
Pro54 (ideal mass)	5.079	6.069	3000.0	23.32	301.0	137.7	0.89	36.6
Pro54 (ideal drag)	6.530	7.520	2999.9	25.04	251.8	109.2	0.75	32.5
Pro75 (ideal mass)	9.800	11.872	3000.0	25.85	250.1	58.3	0.74	21.9
Pro75 (ideal drag)	7.905	9.977	3000.1	24.31	286.9	69.8	0.85	24.2

(a) Altitude vs. time. (b) Velocity vs. time.

Figure 2: Simulated flight profiles for the two configurations.

TODO: add altitude/velocity plots and confirm subfigure captions for figure 2 once StabTraj simulations are ready

2.2 Systems

Our project is organized into different systems based on the area of the rocket being worked on.

2.2.1 Air frame

The air frame team's goal is to design the outer surface of the rocket. This team is split into three sub-teams: aerodynamics, stability, and structural.

The aerodynamics sub-team's goal is to create and optimize surfaces, and to extract drag, lift, and pitching-moment data. Part of their current work is shown in subfigure 3a.

The stability sub-team derives the analytical models for drag and lift (as functions of Mach number and angle of attack), and derives and evaluates the rocket's stability using these coefficients.

The structural sub-team's goal is to define and optimize the dimensions and features of the solid parts using FEA. Their work is shown in subfigure 3b.

The list of structural components is: nosecone, main pipe, transition, motor pipe, tailcone, fins, and rail button.

This year we want to manufacture a detachable rail guide to minimize aerodynamic imbalance; an example can be seen in this video.

(a) Surface assembly. (b) Solid assembly.

Figure 3: Current air frame design work.

2.2.2 Air brake

The air brake team is currently in the design stage. Based on reports from other projects, we have selected and adapted one of the existing air-brake extension mechanisms, shown in figure 4.

The team is organized into eight parts: design (CAD), drivetrain mechanics (motor and gear sizing), air-brake mechanics (aerodynamic and friction force evaluation), structural (FEA), aerodynamics (CFD), opening-controller design, and trajectory-controller design.

Figure 4: Selected air-brake extension mechanism.

The desired trajectory depends on the motor assigned to us and on Monte Carlo trajectory simulations. We are currently aiming for an apogee of around 2250 m with the air brakes deployed.

For the opening and trajectory controllers, we plan to evaluate three methods, from simplest to most complex: PD control, bounding-trajectory blending, and MPC, using our in-house simulator. We also plan to implement a Kalman filter for state estimation.

2.2.3 Structural

The structural team deals with every structural component that the air frame team does not work on.

The team's goal is to derive the free-body diagram of each component, work out the magnitude and direction of the reaction and applied forces, use FEA to find the minimum mass for each loading configuration, and use topology optimization to further reduce the mass.

The list of structures involved is: bulkhead, transition rings and coupler, parachute bay, electronics bay, reaction-wheel plates, and guide rail.

As part of the structural team's work, we plan to develop a reusable inner guide system that lets us slide components in and out, since the rocket's small diameter makes it difficult to work by hand and align parts with screws.

2.2.4 Reaction wheel

This is an experiment we have attempted twice over the past three years. We already have a certain amount of background knowledge, so this would not be something entirely new.

The team is split into: reaction-wheel design, equations of motion, and controller design.

2.2.5 Parachute opening mechanism

This year we are choosing between a conventional design, a sliding door, and nosecone parachute ejection.

The parachute team is split into: door-opening mechanism, parachute ejection mechanism, and parachute bay design.

2.3 Manufacturing

The pipes will be manufactured from carbon fiber, and the nosecone, transition, and tailcone from glass fiber. The fins are going to be manufactured using carbon fiber and reinforced with a tip-to-tip layup.

The air brake, structural, reaction wheel, and parachute opening mechanism components are going to be manufactured by machining aluminium plates and X2C material from one of our sponsors.

Majority of parts will be tested prior with 3D printing and assembled on PVC pipe.

2.4 Tests

The tests are performed to verify the results obtained from tools such as FEA and CFD. For structural, we are planning on testing against the C'Space rules using our own test rings, and we are also planning on doing our own tests (with the university) to verify the models. This includes testing the structure's natural frequencies using a shaker provided by the LaMCoS laboratory. For CFD, we have contact with the wind tunnel, where we will verify the rocket's and air brake's performance.

3 Electrical

4 Team structure

4.1 About

Our team is mainly composed of INSA Lyon students, with the exception of students from

- INSA Strasbourg
- TODO: Julie IUT Lyon 1
- TODO: Oliver.

The teams operate in English, due to the significant number of international members.

4.2 Organization

Our club's main leaders are Paul Retourné (FX27 lead), Yaroslav Knyazkov (Mechanical lead), and Issam Kerrouche (Electrical lead).

The Fusex project is split into an electrical team and a mechanical team, with the mechanical team further subdivided into systems. Each team is run by a team lead, and team meetings are held weekly. Tasks are tracked in a shared spreadsheet and divided by team, as shown in figure 5. We are in the process of developing a spreadsheet for variable management and an inquiry system, where one team can request a parameter from another team.

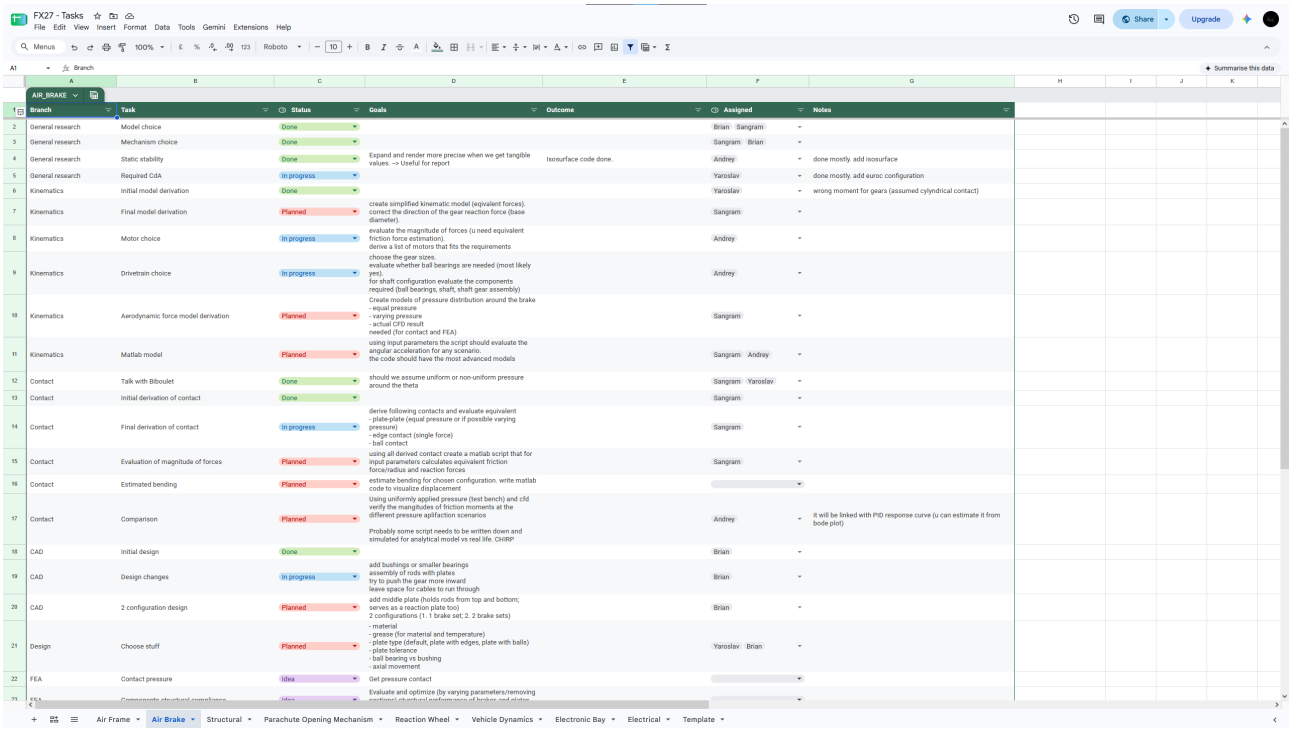


Figure 5: Team structure and task breakdown.

5 Questions

1. Notre conception actuelle utilise un cône arrière (*tail cone*) composé de deux parties : (1) un cône arrière Pro54 du fabricant, et (2) une surface supplémentaire placée dessus afin qu'il soit au ras du tube de la fusée. Nous comprenons que les cônes arrière ne sont pas fournis par le C'Space. Nous en achèterons un pour une autre compétition, mais nous aimerions savoir si nous pouvons utiliser cette conception au C'Space, ou si elle n'est pas autorisée.
2. Nous croyons nous souvenir qu'au C'Space, il est interdit de larguer quoi que ce soit depuis la fusée (sauf si l'élément reste attaché), pour des raisons de sécurité et de protection de l'environnement. Cette année, nous prévoyons de développer un bouton de rail qui se détache pendant le lancement. Cela serait-il acceptable ? Notre conception s'inspire de cette vidéo : https://youtu.be/LnunMEopsb8?t=83
3. If we are allocated the Pro75 motor, would we be able to switch to the Pro54 if a design change comes up later? This is unlikely, but we would like to know just in case.
4. What are the rail dimensions? We have only been able to find the maximum rocket dimensions, not the rail size, which would be useful for the rocket's vehicle dynamics.
5. Regarding the 3000 m apogee limit, we understand it is a maximum. Should we aim slightly lower, e.g. within 100 m of the 3000 m limit? Or is it acceptable as long as StabTraj predicts less than 3000 m?
6. If we fly transonic, would it be acceptable to still use composite fins provided we demonstrate their strength? We are in contact with another team using the same method we intend to use; their predicted maximum loading is 35 kg per fin, which should be sufficient.